

SIG787 -Mathematics for AI
Assignment 2
Trimester 1, 2024
Due: Sunday 19 May 2024, 11:59 pm IST

Important notes:

- Your submission has to be handwritten but it must be legible. If your submission is not legible, it will not be marked, resulting in a zero mark. A proper way of presenting your solutions is part of the assessment.
- Please follow the order of questions in your submission.
- All steps (workings) to arrive at the answer must be clearly shown. No marks will be awarded for answers without workings.
- Generally, you need to keep your answers in the form of quotients and surds (e.g. $\frac{2}{3}$ and $\sqrt{3}$). Rarely, you may convert your solutions into decimals for plotting or comparing purposes. However, you need to show the final answer in terms of quotients and surds before converting them into decimals.
- Only (scanned) electronic submission would be accepted via the Olympus.
- Your submission must be in ONE pdf file. Multiple files and/or in different file formats, e.g. .jpg, will NOT be accepted. It is your responsibility to ensure your file is not corrupted and can be read by a standard PDF viewer. Failure to comply will result in a zero mark.

Question 1) Consider these vectors:

$$\mathbf{u} = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 1 \end{bmatrix}, \mathbf{v} = \begin{bmatrix} 0 \\ 2 \\ 1 \\ 1 \end{bmatrix}, \text{ and } \mathbf{w} = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 3 \end{bmatrix}$$

(i) Determine which two vectors are most similar to each other based on these norms:

(a) ℓ_2 norm:

$$\text{dist}(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|_2 = \|\mathbf{y} - \mathbf{x}\|_2 = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

(b) ℓ_1 norm:

$$\text{dist}(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|_1 = \|\mathbf{y} - \mathbf{x}\|_1 = \sum_{i=1}^n |x_i - y_i|$$

(ii) Determine which two vectors are most similar to each other based on the cosine similarity measure:

$$\cos(\theta) = \frac{\mathbf{x} \cdot \mathbf{y}}{\|\mathbf{x}\| \|\mathbf{y}\|}$$

[You can use decimals here. However, you need to show the final answer in terms of quotients and surds before converting them into decimals.]

- (iii) Explain the reason behind the difference in result between (i) and (ii) if you observe any.
- (iv) Can the difference be resolved? Give details of your suggestion, if you have any, and explain the outcome if your suggestions are applied.

[5+5+5+5= 20 marks]

Question 2) You are tasked with uncovering information about an incomplete matrix, some of whose entries are unknown and denoted as a, b, c , and d :

$$A = \begin{bmatrix} -1 & 0 & a \\ b & 4 & c \\ d & 0 & 0 \end{bmatrix}$$

- (i) Find the rank of A based on the values a, b, c , and d .
- (ii) Using the results you obtained in (i), how many distinct eigenvalues does A have considering a, b, c , and d ?

[5 + 10 = 15 marks]

Question 3) Consider the following matrix.

$$A = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}$$

- (i) Find the full solution set for $A\mathbf{x} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $\mathbf{x} \in \mathbb{R}^3$.
- (ii) Find $\text{rank}(A)$, a basis for the column space of A , $C(A)$, a basis for the row space of A , $C(A^T)$, a basis for the null space of A , $N(A)$, and a basis for the left null space of A , $N(A^T)$.

Hint: For a $m \times n$ matrix A , the column space of A is defined as $C(A) = \{A\mathbf{x} | \mathbf{x} \in \mathbb{R}^n\} \subseteq \mathbb{R}^m$, and the row space of A is defined as $C(A^T) = \{A^T\mathbf{y} | \mathbf{y} \in \mathbb{R}^m\} \subseteq \mathbb{R}^n$. Also, the nullspace $N(A)$ is defined as $\{\mathbf{x} \in \mathbb{R}^n | A\mathbf{x} = \mathbf{0}\} \subseteq \mathbb{R}^n$. For the same matrix, the left nullspace $N(A^T)$ is defined as $\{\mathbf{y} \in \mathbb{R}^m | A^T\mathbf{y} = \mathbf{0}\} \subseteq \mathbb{R}^m$. Also, we have

$$\dim(C(A)) + \dim(N(A^T)) = m$$

$$\dim(C(A^T)) + \dim(N(A)) = n.$$

- (iii) Construct $B = AA^T$, and find eigenvalues and eigenvectors of B . For positive eigenvalues of B , define $\sigma_i = \sqrt{\lambda_i}$, where $\lambda_1 \geq \lambda_2$. Construct matrix D as follows.

$$D = \begin{bmatrix} \sigma_1 & 0 & 0 \\ 0 & \sigma_2 & 0 \end{bmatrix}$$

- (iv) If the eigenvectors of B are not orthonormal, orthonormalise them. Make a matrix U using the orthonormal vectors you obtained. The ordering of the columns of U should be the same as the ordering of the eigenvalues, that is $U = [\mathbf{u}_1 \ \mathbf{u}_2]$. In other words, $\mathbf{u}_1$ is the eigenvector corresponding λ_1 and $\mathbf{u}_2$ is the eigenvector corresponding to λ_2 . Show that U is an orthogonal matrix.
- (v) Find eigenvalues and eigenvectors of the matrix $G = A^T A$, and orthonormalise them. Call them $\mathbf{v}_1, \mathbf{v}_2$, and $\mathbf{v}_3$ according to the order of $\lambda_1 \geq \lambda_2 \geq \lambda_3$ of eigenvalues of G .
- (vi) Orthogonalise $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$, and construct the matrix $V = [\mathbf{v}_1 \ \mathbf{v}_2 \ \mathbf{v}_3]$, and check whether it is an orthogonal matrix.
- (vii) Find three vectors $\{\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3\}$ so that

$$\mathbf{w}_i = \frac{1}{\sigma_i} A^T \mathbf{u}_i, i = 1, 2$$

$$\mathbf{w}_3 \perp \mathbf{w}_1, \mathbf{w}_3 \perp \mathbf{w}_2, \text{ and } \|\mathbf{w}_3\| = 1.$$

Compare $\{\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3\}$ with $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ and comment similar characteristic of these sets.

- (viii) Compute the product UDV^T .
- (ix) Compare the columns of U and V with the bases you found for $C(A), C(A^T), N(A)$, and $N(A^T)$. What can you say about the columns of U and V in terms of the bases for the four subspaces of the matrix A ?

[5+10+5+5+10+5+10+5+10= 65 marks]